

- Description

This module manages a smart fertilizer dispenser for plants, ensuring that:

1. The system continuously monitors the soil nutrient level (as a percentage).
2. If the nutrient level falls below a predefined minimum threshold, the dispenser activates to release fertilizer.
3. If the nutrient level rises above a predefined maximum threshold, the dispenser remains off to prevent over-fertilization.
4. When the nutrient level is within the optimal range, no fertilizer is dispensed.

This system helps optimize plant growth by maintaining an ideal soil nutrient balance and preventing under- or over-fertilization.

- Specification

module SmartFertilizerDispenserControl

var currentNutrientLevel, minNutrientLevel, maxNutrientLevel: real

var dispensingActive: boolean

process updateNutrientLevel(newNutrientLevel: real) returns (currentNutrientLevel: real, dispensingActive: boolean)

functional scenario form:

currentNutrientLevel < minNutrientLevel && dispensingActive = true, Fertilizer dispenser activated.

||

currentNutrientLevel > maxNutrientLevel && dispensingActive = false, Fertilizer dispenser deactivated.

||

currentNutrientLevel >= minNutrientLevel && dispensingActive = false, Dispenser remains off.

end process

end module

- Java Code

```
package code;
public class c82_SmartFertilizerDispenser {
    private double minNutrientLevel;
    private double maxNutrientLevel;
    private double currentNutrientLevel;
    private boolean dispensingActive;
    public c82_SmartFertilizerDispenser(double minNutrientLevel, double
maxNutrientLevel, double initialNutrientLevel) {
        this.minNutrientLevel = minNutrientLevel;
        this.maxNutrientLevel = maxNutrientLevel;
        this.currentNutrientLevel = initialNutrientLevel;
        this.dispensingActive = false;
    }
}
```

```

    }
    public void updateNutrientLevel(double newNutrientLevel) {
        currentNutrientLevel = newNutrientLevel;
        adjustFertilizerDispensing();
        System.out.printf("Current Nutrient Level: %.2f%%, Fertilizer Dispensing: %s\n",
                           currentNutrientLevel,
                           dispensingActive ? "ON" : "OFF");
        if (dispensingActive) {
            System.out.println("ALERT: Low nutrient level detected. Activating fertilizer dispenser.");
        } else if (currentNutrientLevel > maxNutrientLevel) {
            System.out.println("INFO: Nutrient level is high. Fertilizer dispensing is deactivated.");
        } else {
            System.out.println("Nutrient level is optimal. No action needed.");
        }
    }
    private void adjustFertilizerDispensing() {
        if (currentNutrientLevel < minNutrientLevel) {
            dispensingActive = true;
        } else if (currentNutrientLevel > maxNutrientLevel) {
            dispensingActive = false;
        } else {
            dispensingActive = false;
        }
    }
    public double getCurrentNutrientLevel() {
        return currentNutrientLevel;
    }

    public boolean isDispensingActive() {
        return dispensingActive;
    }

    public static void main(String[] args) {
        c82_SmartFertilizerDispenser dispenser = new c82_SmartFertilizerDispenser(40.0, 70.0, 55.0);

        // testcase-VT:
        //     dispenser.updateNutrientLevel(35.0);
        //     dispenser.updateNutrientLevel(65.0);
        //     dispenser.updateNutrientLevel(75.0);

```

```

        // testcase-FT:
//        dispenser.updateNutrientLevel(0.0);
//        dispenser.updateNutrientLevel(1.7976931348623157E308);
//        dispenser.updateNutrientLevel(-5.005342255597914E307);

        // testcase-Z3:
//        dispenser.updateNutrientLevel(35.0);
//        dispenser.updateNutrientLevel(65.0);
//        dispenser.updateNutrientLevel(75.0);

        // testcase-UVT:
        dispenser.updateNutrientLevel(35.0);
        dispenser.updateNutrientLevel(65.0);
        dispenser.updateNutrientLevel(75.0);
    }
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	1.2 %	10	832	842
src	1.2 %	10	832	842
code	1.2 %	10	832	842
c82_SmartFertilizerDispenser.java	100.0 %	10	0	10
c82_SmartFertilizerDispenser	100.0 %	10	0	10
main(String[])		0	0	0
c82_SmartFertilizerDispenser(do		0	0	0
adjustFertilizerDispensing()	100.0 %	4	0	4
updateNutrientLevel(double)	100.0 %	6	0	6

3. Vibration testing instruction coverage: 94.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.5 %	103	21,057	21,160
src	0.5 %	103	21,057	21,160
code	0.5 %	103	21,057	21,160
c82_SmartFertilizerDispenser.java	94.5 %	103	6	109
c82_SmartFertilizerDispenser	94.5 %	103	6	109
main(String[])	100.0 %	17	0	17
c82_SmartFertilizerDispenser(do	100.0 %	15	0	15
adjustFertilizerDispensing()	100.0 %	24	0	24
updateNutrientLevel(double)	100.0 %	47	0	47

4. Fuzz testing branch coverage: 80%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	1.0 %	8	834	842
src	1.0 %	8	834	842
code	1.0 %	8	834	842
c82_SmartFertilizerDispenser.java	80.0 %	8	2	10
c82_SmartFertilizerDispenser	80.0 %	8	2	10
main(String[])		0	0	0
c82_SmartFertilizerDispenser(do		0	0	0
adjustFertilizerDispensing()	75.0 %	3	1	4
updateNutrientLevel(double)	83.3 %	5	1	6

5. Fuzz testing instruction coverage: 89%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	97	21,063	21,160
▼ src	0.5 %	97	21,063	21,160
▼ code	0.5 %	97	21,063	21,160
▼ c82_SmartFertilizerDispenser.java	89.0 %	97	12	109
▼ c82_SmartFertilizerDispenser	89.0 %	97	12	109
main(String[])	100.0 %	17	0	17
c82_SmartFertilizerDispenser(do	100.0 %	15	0	15
adjustFertilizerDispensing()	87.5 %	21	3	24
updateNutrientLevel(double)	93.6 %	44	3	47

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.2 %	10	830	840
▼ src	1.2 %	10	830	840
▼ code	1.2 %	10	830	840
▼ c82_SmartFertilizerDispenser.java	100.0 %	10	0	10
▼ c82_SmartFertilizerDispenser	100.0 %	10	0	10
adjustFertilizerDispensing()	100.0 %	4	0	4
updateNutrientLevel(double)	100.0 %	6	0	6
main(String[])		0	0	0
c82_SmartFertilizerDispenser(double		0	0	0

7. Z3 instruction coverage: 94.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	103	21,057	21,160
▼ src	0.5 %	103	21,057	21,160
▼ code	0.5 %	103	21,057	21,160
▼ c82_SmartFertilizerDispenser.java	94.5 %	103	6	109
▼ c82_SmartFertilizerDispenser	94.5 %	103	6	109
main(String[])	100.0 %	17	0	17
c82_SmartFertilizerDispenser(do	100.0 %	15	0	15
adjustFertilizerDispensing()	100.0 %	24	0	24
updateNutrientLevel(double)	100.0 %	47	0	47

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.2 %	10	830	840
▼ src	1.2 %	10	830	840
▼ code	1.2 %	10	830	840
▼ c82_SmartFertilizerDispenser.java	100.0 %	10	0	10
▼ c82_SmartFertilizerDispenser	100.0 %	10	0	10
adjustFertilizerDispensing()	100.0 %	4	0	4
updateNutrientLevel(double)	100.0 %	6	0	6
main(String[])		0	0	0
c82_SmartFertilizerDispenser(double		0	0	0

9. Unimproved V-method instruction coverage: 94.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	103	21,057	21,160
▼ src	0.5 %	103	21,057	21,160
▼ code	0.5 %	103	21,057	21,160
▼ c82_SmartFertilizerDispenser.java	94.5 %	103	6	109
▼ c82_SmartFertilizerDispenser	94.5 %	103	6	109
main(String[])	100.0 %	17	0	17
c82_SmartFertilizerDispenser(do	100.0 %	15	0	15
adjustFertilizerDispensing()	100.0 %	24	0	24
updateNutrientLevel(double)	100.0 %	47	0	47